

Week 3 - Day 3: Classification & Logistic Regression

5-Minute Review Notes

Day Goal

Understand classification, Logistic Regression, class probabilities, thresholds, confusion matrices, accuracy, class imbalance, and why a perfect score on a tiny test set does not prove real-world performance.

1. Regression vs Classification

Regression	Classification
Predicts a continuous numerical value	Predicts a category/class
Example: Salary, House Price	Example: Approved/Rejected, Spam/Not Spam
LinearRegression	LogisticRegression

Important: class labels stored as 0 and 1 are still categories. The meaning of the target determines whether a problem is regression or classification.

2. Binary vs Multiclass Classification

Binary: exactly two classes, such as Approved/Rejected or Churn/Stay. **Multiclass:** more than two classes, such as Low/Medium/High/Critical.

3. Logistic Regression

Linear Regression can predict unrestricted values such as -0.4 or 1.35. Logistic Regression models class probabilities and assigns a class using a decision threshold.

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

4. Probability and Threshold

A common default threshold is 0.5: probability of class 1 ≥ 0.5 usually becomes class 1; otherwise class 0.

P(Class 1)	Prediction
0.20	Class 0
0.49	Class 0
0.51	Class 1
0.82	Class 1

5. predict() vs predict_proba()

predict() returns final class labels. **predict_proba()** returns the probability for each class.

```
y_pred = model.predict(X_test)
probs = model.predict_proba(X_test)
y_prob_class_1 = probs[:, 1]
```

6. Confusion Matrix

```
[[TN, FP],  
 [FN, TP]]
```

Term	Meaning	Loan Example
TP	Actual positive, predicted positive	Approved -> Approved
TN	Actual negative, predicted negative	Rejected -> Rejected
FP	Actual negative, predicted positive	Rejected -> Approved
FN	Actual positive, predicted negative	Approved -> Rejected

Memory rule: the second word tells you what the model predicted. False Positive = predicted positive but wrong. False Negative = predicted negative but wrong.

7. Accuracy

Accuracy = Correct Predictions / Total Predictions. Correct predictions are TP + TN.

```
accuracy = accuracy_score(y_test, y_pred)
```

8. Why Accuracy Can Mislead

If 990 of 1000 transactions are legitimate and 10 are fraud, a useless model that predicts everything as legitimate still gets 99% accuracy while detecting 0 of 10 fraud cases. This is why class imbalance matters.

9. stratify=y

stratify=y tries to preserve approximately the same target-class proportions in the training and testing sets.

```
train_test_split(  
    X, y, test_size=0.20, random_state=42, stratify=y  
)
```

10. Day 3 Practical - Loan Approval

Features: Income, Credit_Score, Debt. Target: Loan_Approved (0 = Rejected, 1 = Approved). Dataset size: 30 rows.

Class distribution: 19 approved (about 63%) and 11 rejected (about 37%). Somewhat imbalanced, but not severely imbalanced.

Metric / Result	Value
Test size	6 rows
Accuracy	1.0 (100%)
Confusion Matrix	[[2, 0], [0, 4]]
TN / FP / FN / TP	2 / 0 / 0 / 4

All 6 test examples were classified correctly. This does not prove the model is perfect: the dataset is tiny, the test set has only 6 rows, and the pattern is intentionally simple. Real-world performance requires much more unseen and realistic data.

11. Key Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

X = df[["Income", "Credit_Score", "Debt"]]
y = df["Loan_Approved"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)
probs = model.predict_proba(X_test)
```

12. Assessment - Quick Answers

- Q1.** 0 and 1 represent categories, so loan approval is classification.
- Q2.** Binary has two classes; multiclass has more than two.
- Q3.** Logistic Regression models class probabilities; Linear Regression predicts continuous values.
- Q4.** predict() gives classes; predict_proba() gives class probabilities.
- Q5.** 0.82 with threshold 0.5 gives class 1.
- Q6.** FP: Rejected but predicted Approved. FN: Approved but predicted Rejected.
- Q7.** [[15,3],[2,20]] -> TN=15, FP=3, FN=2, TP=20; correct=35.
- Q8.** High accuracy can hide complete failure on a minority class.
- Q9.** stratify=y preserves approximate class proportions in the split.

13. 5-Minute Cheat Sheet

Concept	Remember
Classification	Predict a category/class
Logistic Regression	Classification algorithm
predict()	Final class
predict_proba()	Probability for each class
Threshold	Common default: 0.5
Confusion Matrix	[[TN, FP], [FN, TP]]
Accuracy	(TP + TN) / Total
FP	Predicted positive, but wrong
FN	Predicted negative, but wrong
Class imbalance	Accuracy can be misleading
stratify=y	Preserve class proportions

Status: Week 3 - Day 3 COMPLETE

Progress: Week 1 7/7 | Week 2 7/7 | Week 3 3/7